• EQUAZIONI DIFF. DERIV PARZ.

→ LINEARE ↗ 1° ORDINE $\partial_t u + a\,\partial_x u = f$

↘ 2° ORDINE

→ $A\,u_{xx} + B\,u_{xy} + C\,u_{yy} + D\,u_x + E\,u_y + F\,u - g = 0$

IPER $\Delta > 0$

ELL $\Delta < 0$

PARA $\Delta = 0$

$\Delta = B^2 - 4AC$

? È Δ INVARIANTE PER TRASFORMAZIONE

$(x, y) \to (\xi, \eta)$

$\alpha^{-1} := \frac{\partial x}{\partial \xi}$ $\quad \beta^{-1} := \frac{\partial x}{\partial \eta}$ $\quad \gamma^{-1} := \frac{\partial y}{\partial \xi}$ $\quad \delta^{-1} := \frac{\partial y}{\partial \eta}$ ||

$\boxed{u_{xx}} = (u_x)_x = (u_x)_\eta \cdot \frac{d\eta}{dx} + (u_x)_\xi \frac{d\xi}{dx} = (u_x)_\eta \cdot \beta + (u_x)_\xi \cdot \alpha = *$

→ $u_x = u_\eta\,\beta + u_\xi \cdot \alpha$ $\qquad * = \beta^2 u_{\eta\eta} + 2\alpha\beta\, u_{\xi\eta} + \alpha^2 u_{\xi\xi}$

$\frac{d}{dx}\left(\frac{du}{dx}\right) = \frac{d}{dx}\left(\frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta}\frac{\partial \eta}{\partial x}\right) =$

$u_{xy} = (u_\eta \beta + u_\xi \alpha)_y = \beta\left(u_{\eta\eta}\frac{\partial \eta}{\partial y} + u_{\eta\xi}\frac{\partial \xi}{\partial y}\right) + \alpha\left(u_{\xi\xi}\frac{\partial \xi}{\partial y} + u_{\xi\eta}\frac{\partial \eta}{\partial y}\right)$

$= \beta\delta\, u_{\eta\eta} + \beta\gamma\, u_{\eta\xi} + \alpha\gamma\, u_{\xi\xi} + \alpha\delta\, u_{\eta\xi}$

$= \beta\delta\, u_{\eta\eta} + (\beta\gamma + \alpha\delta)\, u_{\eta\xi} + \alpha\gamma\, u_{\xi\xi}$

$u_{yy} = \delta^2 u_{\eta\eta} + 2\gamma\delta\, u_{\eta\xi} + \gamma^2 u_{\xi\xi}$

$A\beta^2 u_{\eta\eta} + 2A\alpha\beta\, u_{\xi\eta} + A\alpha^2 u_{\xi\xi}$

$+ B\beta\delta\, u_{\eta\eta} + B(\beta\gamma + \alpha\delta)\, u_{\eta\xi} + B\alpha\gamma\, u_{\xi\xi}$

$+ C\delta^2 u_{\eta\eta} + 2C\gamma\delta\, u_{\eta\xi} + C\gamma^2 u_{\xi\xi}$ + TERMINI 1° ORDINE $= 0$

$$\underbrace{\left(A\beta^2 + B\beta\delta + C\delta^2\right)}_{=:A'} u_{\xi\xi} +$$

$$\underbrace{\left(2A\alpha\beta + B(\beta\gamma + \alpha\delta) + 2C\gamma\delta\right)}_{=:B'} u_{\xi\eta} +$$

$$\underbrace{\left(A\alpha^2 + B\alpha\gamma + C\gamma^2\right)}_{=:C'} u_{\eta\eta} + \dots = 0$$

$$\Delta' = B'^2 - 4A'C'$$

$$= \cancel{4A^2\alpha^2\beta^2} + B^2(\beta^2\gamma^2 + 2\alpha\gamma\beta\delta + \alpha^2\delta^2) + \cancel{4C^2\gamma^2\delta^2}$$

$$+ 4AB\alpha\beta(\beta\gamma + \alpha\delta) + 8AC\alpha\beta\gamma\delta + 4BC(\beta\gamma + \alpha\delta)\gamma\delta$$

$$-4\Big[\cancel{A^2\alpha^2\beta^2} + AB\alpha\beta^2\gamma + AC\beta^2\gamma^2 + AB\alpha^2\beta\delta + B^2\alpha\beta\gamma\delta$$

$$+ BC\beta\delta\gamma^2 + AC\alpha^2\delta^2 + BC\alpha\gamma\delta^2 + \cancel{C^2\gamma^2\delta^2}\Big)$$

$$= AB\left(\cancel{4\alpha\beta^2\gamma} + \cancel{4\alpha^2\beta\delta} - \cancel{4\alpha\beta^2\gamma} - \cancel{4\alpha^2\beta\delta}\right)$$

$$+ AC\left(8\alpha\beta\gamma\delta - 4\beta^2\gamma^2 - 4\alpha^2\delta^2\right)$$

$$+ BC\left(\cancel{4\beta\gamma^2\delta} + \cancel{4\alpha\gamma\delta^2} - \cancel{4\beta\delta\gamma^2} - \cancel{4\alpha\gamma\delta^2}\right)$$

$$+ B^2\left(\beta^2\gamma^2 + \alpha^2\delta^2 + 2\alpha\beta\gamma\delta - 4\alpha\beta\gamma\delta\right)\Big] = B^2\left(\beta^2\gamma^2 + \alpha^2\delta^2 - 2\alpha\beta\gamma\delta\right)$$

$$= AC\left[-4(\alpha\delta - \beta\gamma)^2\right] + B^2(\alpha\delta - \beta\gamma)^2 \qquad \Big|\ J = \alpha\delta - \beta\gamma$$

$$= J^2\left[B^2 - 4AC\right]$$

$$= \underbrace{J^2}_{>0} \cdot \Delta$$

$$J = \begin{vmatrix} \alpha & \beta \\ \gamma & \delta \end{vmatrix} = \begin{vmatrix} \frac{\partial \xi}{\partial x} & \frac{\partial \eta}{\partial x} \\ \frac{\partial \xi}{\partial y} & \frac{\partial \eta}{\partial y} \end{vmatrix} \boxed{\neq 0}$$

$$\operatorname{sign}(\Delta') = \operatorname{sign}(\Delta) = \begin{cases} >0 & \Delta > 0 \\ =0 & \Delta = 0 \\ <0 & \Delta < 0 \end{cases}$$

OBIETTIVO:

- EQUAZIONE IPERBOLICA CANONICA

ES: EQ. ONDE $\partial_{tt} u - c^2 \partial_{xx} u = 0$

$$\Delta = 0 + 4c^2 > 0 \qquad c > 0$$

$$\boxed{\partial_{\eta\xi} v = 0}$$

$\Rightarrow \eta = x + ct \qquad \xi = x - ct \qquad \eta_x = 1 \qquad \eta_t = c$

$\Rightarrow \xi_x = 1 \qquad \xi_t = -c$

$$\partial_{tt} u - c^2 \partial_{xx} u = \left(\left(\frac{\partial \xi}{\partial t}\right)^2 u_{\xi\xi} + 2\left(\frac{\partial \xi}{\partial t}\right)\left(\frac{\partial \eta}{\partial t}\right) \cdot u_{\xi\eta} + \left(\frac{\partial \eta}{\partial t}\right)^2 \cdot u_{\eta\eta} \right)$$

$$- c^2 \left[\left(\left(\frac{\partial \xi}{\partial x}\right)^2 u_{\xi\xi} + 2\left(\frac{\partial \xi}{\partial x}\right)\left(\frac{\partial \eta}{\partial x}\right) u_{\xi\eta} + \left(\frac{\partial \eta}{\partial x}\right)^2 u_{\eta\eta} \right) \right]$$

$$= \cancel{c^2 u_{\xi\xi}} - 2c^2 u_{\xi\eta} + \cancel{c^2 u_{\eta\eta}} - c^2 \left[\cancel{u_{\xi\xi}} + 2 u_{\xi\eta} + \cancel{u_{\eta\eta}} \right]$$

$$= -4c^2 u_{\xi\eta} = 0$$

$$\Rightarrow \boxed{u_{\xi\eta} = 0}$$

$$\int^{\xi} \int^{\eta} \partial_\xi \underbrace{\partial_\eta u} = u(\xi, \eta) \qquad \int_a^b f' = f(b) - f(a)$$

$$\int^{\xi} f(\xi)\, d\xi = \boxed{F(\xi) + G(\eta)}$$

$$0 = \int_{\xi_0}^{\xi} \int_{\eta_0}^{\eta} u_{\xi\eta} = \int_{\xi_0}^{\xi} \Big(\underbrace{u_\xi(\eta, \xi)}_{(1)} - \underbrace{u_\xi(\eta_0, \xi)}_{(2)} \Big)\, d\xi$$

$$= \underbrace{u(\eta, \xi) - u(\eta, \xi_0)}_{(1)} - \underbrace{u(\eta_0, \xi) + u(\eta_0, \xi_0)}_{(2)}$$

② $u(\eta,\xi) = \underbrace{u(\eta,\xi_0)}_{G(\eta):=} + \underbrace{u(\eta_0,\xi) + u(\eta_0,\xi_0)}_{F(\xi):=}$ ←

① $u(\eta,\xi) := G(\eta) + F(\xi)$ ←

$u_{\xi\eta} = \underbrace{G(\eta)_{\xi\eta}}_{=0} + \underbrace{F(\xi)_{\xi\eta}}_{=0} = 0$ ✓ ←

② VERIFICA PDE $\boxed{u_{\xi\eta} = 0}$

② TUTTE LE SOL DELLA PDE SONO NELLA FORMA ①

F, G ARBITRARIE $: \mathbb{R} \to \mathbb{R}$

$$u(\eta,\xi) = F(\xi) + G(\eta) = F(x-ct) + G(x+ct)$$

COME ARRIVARE ALLA FORMA CANONICA?

$$A\,\partial_{xx}u + B\,\partial_{xy}u + C\,\partial_{yy}u = 0 \qquad \Delta > 0$$

$(x,y) \to (\xi,\eta)$: $\qquad \partial_{\xi\eta} v = 0$

$$\overbrace{(A\xi_x^2 + B\xi_x\xi_y + C\xi_y^2)\,u_{\xi\xi}}^{(1)} + \overbrace{(A\eta_x^2 + B\eta_x\eta_y + C\eta_y^2)\,u_{\eta\eta}}^{(2)}$$
$$+ \left[2A\xi_x\eta_x + B(\xi_x\eta_y + \xi_y\eta_x) + 2C\xi_y\eta_x\right]\underbrace{v_{\xi\eta}}_{(3)} = 0$$

① SPARISCA ② SPARISCA ③ RIMANGA

④ $\begin{cases} A\xi_x^2 + B\xi_x\xi_y + C\xi_y^2 = 0 & \leftarrow \xi(x,y) : \\ A\eta_x^2 + B\eta_x\eta_y + C\eta_y^2 = 0 & \leftarrow \eta(x,y) \end{cases}$

$y(x)$: $\qquad \xi(x, y(x)) = \text{costante}$

$$0 = \frac{d\xi}{dx} = \frac{\partial\xi}{\partial x} + \frac{\partial\xi}{\partial y}\cdot\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = -\frac{\frac{\partial\xi}{\partial x}}{\partial\xi/\partial y} = -\frac{\xi_x}{\xi_y}$$

④ $\dfrac{}{\xi_y^2}$

$$A\left(\frac{\xi_x}{\xi_y}\right)^2 + B\frac{\xi_x}{\xi_y} + C = 0$$

$$A\left(\frac{dy}{dx}\right)^2 - B\frac{dy}{dx} + C = 0$$

$$\Delta = B^2 - 4AC > 0$$

$$\left(\frac{dy}{dx}\right) = \begin{cases} \dfrac{B-\sqrt{\Delta}}{2A} \rightsquigarrow y(x) = \left(\dfrac{B-\sqrt{\Delta}}{2A}\right)x + \xi_0 \\[2ex] \dfrac{B+\sqrt{\Delta}}{2A} \rightsquigarrow y(x) = \left(\dfrac{B+\sqrt{\Delta}}{2A}\right)x + \eta_0 \end{cases}$$

$\rightsquigarrow$ $\xi(x,y)$, $\eta(x,y)$ LINEARMENTE INDIPENDENTI

$\Rightarrow$ SI FANNO I CONTI ... $\rightarrow$ (1) e (2) SPARISCONO

e (3) DIVENTA QUALCOSA $\neq 0$ $\boxed{v_{\eta\eta} = 0}$

ES. $\boxed{u_{tt} + u_{tx} = 0}$ $\qquad (t,x) \to (\xi,\eta)$

$$0 = \underbrace{(\xi_t^2 + \xi_t\xi_x)}_{A} u_{\xi\xi} + \underbrace{(2\xi_t\eta_t + \xi_t\eta_x + \xi_x\eta_t)}_{B} u_{\xi\eta} + \underbrace{(\eta_t^2 + \eta_t\eta_x)}_{C} u_{\eta\eta}$$

$A = 0$, $C = 0$ $\qquad \begin{cases} \xi_t^2 + \xi_t\xi_x = 0 \\ \eta_t^2 + \eta_t\eta_x = 0 \end{cases}$

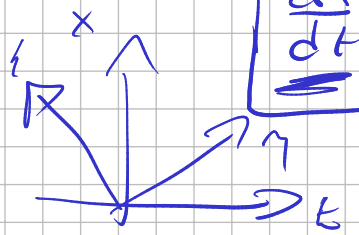
$x(t)$: $\xi(t, x(t)) = \xi_0$

$$\boxed{\frac{\partial\xi}{\partial t} + \frac{\partial\xi}{\partial x}\frac{dx}{dt} = 0}$$

$$\rightarrow \left(\frac{\xi_t}{\xi_x}\right)^2 + \frac{\xi_t}{\xi_x} = 0$$

$$\rightarrow \left(\frac{dx}{dt}\right)^2 - \frac{dx}{dt} = 0$$

$$\rightarrow \boxed{\frac{dx}{dt} = -\frac{\xi_t}{\xi_x}}$$



RETTE CARATTERISTICHE

$$\left(\frac{dx}{dt}\right)\left(\frac{dx}{dt}-1\right)=0$$

TROVO NUOVE COORDINATE

① $\frac{dx}{dt}=0$ $\quad x(t)=\xi_0$

② $\frac{dx}{dt}=1$ $\quad x(t)=t+\eta_0$

$\Rightarrow$ $\xi(x,t)=x$, $\eta(x,t)=x-t$

- PARABOLICHE FORMA CANONICA? $\underline{\Delta=0}$

$\Delta=0$ $\qquad \frac{dx}{dt}=\frac{B\pm\sqrt{\Delta}}{2A}=\frac{B}{2A}$

1 ROTTA CARATTERISTICA $\Rightarrow$ 1 VARIABILE

$\boxed{\eta=2Ay-Bx}$ $\rightarrow C'=0$ $\qquad \underline{\xi=x}$

$A'\neq 0$

$0=\Delta'=B'^2-4A'C'=B'^2$

$\Rightarrow B'=0$

$\Rightarrow A'v_{\xi\xi}=0$ $\qquad v(\xi,\eta)=F(\eta)+\xi G(\eta)$

$F, G: \mathbb{R}\to\mathbb{R}$ ARBITRARIE

$v(\xi,\eta)=F(x)+(2Ax-By)G(x)$

ELLITTICHE $\quad \Delta<0$

NON ESISTONO ROTTE CARATTERISTICHE!

$\frac{dy}{dx}=\frac{B\pm\sqrt{\Delta}}{2A}$ $\qquad \Delta<0$!

$$A' u_{\xi\xi} + B' u_{\xi\eta} + C' u_{\eta\eta} = 0$$

NUOVO OBIETTIVO : $B' = 0$ $\qquad A' = C'$

$$* = \left[2A \xi_x \eta_x + B(\xi_x \eta_y + \xi_y \eta_x) + 2C \underbrace{\xi_y \eta_y}_{=0} \right] = 0$$

$\eta = x \qquad \xi = \dfrac{2Ay - Bx}{\sqrt{-\Delta}}$

$\eta_x = 1 \qquad \xi_x = -\dfrac{B}{\sqrt{-\Delta}} \qquad \xi_y = \dfrac{2A}{\sqrt{-\Delta}}$

$\eta_y = 0$

$$* = 2A\left(-\frac{B}{\sqrt{-\Delta}}\right) \cdot 1 + B \frac{2A}{\sqrt{-\Delta}} = 0$$

$$= \frac{-2AB + 2AB}{\sqrt{-\Delta}} = 0 \; !$$ DERANI

$A' = C' \Rightarrow$

$$\boxed{A\left(v_{\xi\xi} + v_{\eta\eta}\right) = 0}$$